

# Practical Learning Reflection

## Restaurant Tipping Statistical Analytics

**Course/Lab:** Data Science & Applied Statistics

**Project:** End-to-End Statistical Modeling & Web Dashboard Delivery

**Dataset:** Restaurant Tipping Behavior (tips.csv)

### 1. Data Cleaning, Preprocessing & Feature Validation

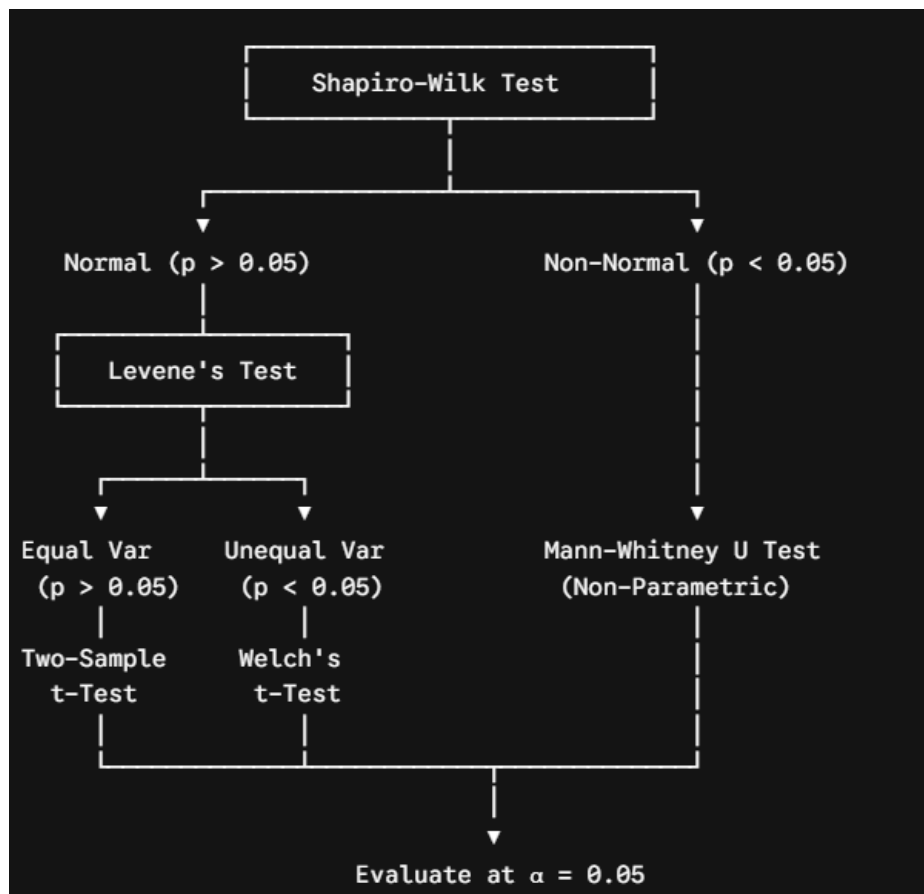
- **Missing Value & Data Integrity Auditing:** Conducted programmatic checks for null values (`df.isnull().sum()`) and duplicate records across features (total\_bill, tip, sex, smoker, day, time, size) to ensure dataset completeness before statistical modeling.
- **Data Type Enforcement & Categorical Encoding:** Verified numerical attributes (total\_bill, tip, size) were stored as floating/integer types and validated categorical attributes (sex, smoker, day, time) to ensure correct factor level assignment for design matrices.
- **Dynamic Slicing vs. Global Model Consistency:** Built Pandas boolean indexing routines to filter data based on user inputs for exploratory visuals, while retaining full categorical factor levels in the underlying dataset to prevent missing-category errors during model prediction.
- **Feature Scope & Boundary Validation:** Computed bounds and verified range constraints across continuous variables (confirming zero-valued or negative tip/bill records were absent) to maintain valid model assumptions.

### 2. Exploratory Data Analysis & Distributional Diagnostics

- **Parametric vs. Non-Parametric Metric Pairing:** Learned that relying strictly on mean and standard deviation can be misleading for skewed financial data like total\_bill and tip. Pairing them with median and Interquartile Range (IQR) provides a robust representation of central tendency and dispersion.
- **Shape Diagnostics:** Evaluated distribution geometry using skewness and kurtosis. Right-skewed distribution characteristics in consumer tipping patterns were confirmed through positive skewness values and visual KDE/histogram overlays.

- **Bivariate Multi-Dimensionality:** Explored non-linear relationships and subgroup trends by mapping numerical metrics against categorical variables using interactive scatter plots color-coded by smoking status and party size.

### 3. Rigorous Inferential Hypothesis Testing



- **Assumption-Driven Test Selection:** Mastered the requirement to systematically verify Gauss-Markov and parametric assumptions before choosing statistical tests rather than assuming normality.
- **Normality Verification:** Utilized the Shapiro-Wilk test to evaluate whether continuous variables met the Gaussian distribution assumption ( $H_0$ : Data is normally distributed).
- **Homogeneity of Variance:** Applied Levene's test to verify equal variance across independent groups ( $H_0$ :  $\sigma_1^2 = \sigma_2^2$ ).
- **Dynamic Decision Engine:** Built logic to automatically route execution to a parametric **Two-Sample t-test** when assumptions hold, or fallback to a non-parametric **Mann-Whitney U**

Test when normality fails.

- **Multi-Group Comparison:** Applied One-Way Analysis of Variance (ANOVA) to evaluate variance across more than two categorical factor levels (e.g., comparing tip averages across days of the week).

## 4. Multiple Linear Regression & Econometric Diagnostics

- **Model Formulation:** Constructed an Ordinary Least Squares (OLS) regression model:  
$$\text{tip} = \beta_0 + \beta_1(\text{total\_bill}) + \beta_2(\text{size}) + \sum \beta_i(\text{Categorical Controls}) + \varepsilon$$
- **Parameter Interpretation:** Evaluated estimated coefficients (Beta /  $\beta$ ), p-values ( $p < 0.05$  for statistical significance), and overall variance explained using  $R^2$  and Adjusted  $R^2$ .
- **Confidence vs. Prediction Intervals:** Differenced between 95% Confidence Intervals (estimating the expected average tip for a group) and 95% Prediction Intervals (estimating the uncertainty of an individual tip prediction).
- **Gauss-Markov Residual Diagnostics:**
  - **Linearity & Homoscedasticity:** Examined Residuals vs. Fitted value scatter plots to confirm random variance scatter around the zero-line without funneling patterns.
  - **Residual Normality:** Generated Q-Q plots and performed Jarque-Bera tests to confirm that model errors ( $\varepsilon$ ) are normally distributed.
  - **Multicollinearity:** Calculated Variance Inflation Factors (VIF) for continuous predictors to ensure predictors were independent ( $\text{VIF} < 5$ ).

## 5. Software Engineering & Production Cloud Deployment

- **Modular Interactive Architecture:** Learned to translate static notebook code into a multi-page interactive web application organized into clear user tabs using Streamlit (`st.tabs`).
- **Session & State Management:** Leveraged dataset caching (`@st.cache_data`) to prevent redundant data reloading and improve application rendering speed.
- **Robust Category Handling:** Handled categorical level retention by fitting global models on full dataset slices to prevent PatsyError matrix mismatches during UI filtering.
- **DevOps Pipeline:** Created clean local development environments using Python virtual environments (`venv`), tracked dependencies using `requirements.txt`, and maintained clean version control using `.gitignore`.

- **Continuous Cloud Deployment:** Successfully deployed a live web application from a public GitHub repository to Streamlit Community Cloud.

### **GitHub Repo Link**

[https://github.com/aarushi-202618017/Stats\\_Lab\\_04\\_Tipping\\_Dashboard](https://github.com/aarushi-202618017/Stats_Lab_04_Tipping_Dashboard)